OO-L-90a
AMENDMENT—1
March 6, 1964

FEDERAL SPECIFICATION

LATHE, BRAKE-DRUM

This Amendment, which forms a part of Federal Specification OO-L-90a, dated September 14, 1962, was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

Page 1, paragraph 1.2.1: Delete "Size I.—Standard-duty.", also delete "Size III.—Extra-heavy-duty." requirements.

Page 1, paragraph 2.1: Delete the paragraph in its entirety and substitute the following new paragraph:

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Pages 1 and 2, paragraph 2.1: Delete the parenthetical source paragraphs following the list of Federal Standards and substitute the following paragraphs:

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards from established distribution points in their agencies.)"

Page 2, paragraph 2.1: Delete Military Specification "MIL-V-173—Varnish, Moisture- and Fungus-Resistant for the Treatment of Communications, Electronic, and Associated Electrical Equipment." and substitute Military Specification "MIL-T-152—Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment."

Page 2, paragraph 2.1: Add the following Military Standard:

"MS 19081—Bearings, Roller, Tapered Single Row of Rollers, Normal Angle."

Page 2, paragraph 2.1: After the list of Military Standards, add the following list of Military Drawings:

"*Drawings, Ordnance:*

D7025887 — Drum Assembly, Wheel Brake

D7328372—Brake Drum and Back Assembly"

Page 2, paragraph 2.1: Delete the final parenthetical source paragraph and substitute the following paragraph:

"(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)"

Page 2, paragraph 2.2: Add to list of American Standards Association Standards:

B6.12-1954 — Nomenclature for Gear Tooth Wear and Failure.

FSC 4910

Page 3, paragraph 2.2: Add the following source paragraph prior to section 3, Requirements:

"(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)"

Page 3, paragraph 3.3.6, line 1: After the word "parts" add "shall be in the inch system and".

Page 4, paragraphs 3.4, 3.4.1, 3.4.1.1 and 3.4.1.2: Delete the paragraphs in their entirety and substitute the following paragraph:

3.4 Performance and product characteristics. The lathe shall be an electric motor driven machine tool capable of truing commercial and military automotive brake drums within the sizes and vehicles classes specified herein. The main drive motor shall provide sufficient power to meet the performance requirements specified. The lathe shall be capable of turning all brake drums 17-1/4 inches diameter or less while mounted as single or dual assemblies not exceeding 1300 pounds or 35 inches maximum overall width (7.00-16 through 14.00-24 tire sizes as listed in the Military Tire and Rim Data Book) without removal of the wheel, tires or tubes from the brake drum and hub. It must be capable of turning larger sizes within specified size range (table I) with the wheels removed. The lathe shall be capable of turning towed vehicle brake drums within the sizes specified in table I.

Page 5, table I: Delete table I in its entirety and substitute the following:

TABLE I.—*Physical characteristics*

Unless otherwise specified (see 6.2), physical characteristics of the size II lathe shall be as follows:

Drum capacity range (dia.), inches (min.)	8.9 to 22
Drum cutting depth (face), inches (min.)	8
Speeds, number of changes (min.)	3
Speeds, cutting, range, r.p.m. (min.)	30 to 100
Swing, inches (min.)	60
Feed, inches per revolution, range (min.)	.005—.016
Number of feeds (min.)	2

Page 5, table II: Delete table II in its entirety.

Page 6, paragraph 3.4.1.3: Delete the paragraph in its entirety.

Page 6, paragraph 3.4.2: Delete the paragraph in its entirety and substitute the following paragraph:

3.4.2 *Machine accuracy.*

Spindle taper run-out—0.0005 inch maximum T.I.R. (Total indicator reading)

Mandrel run-out—0.001 inch maximum T.I.R. nine inches from spindle nose

Adapter taper run-out—0.001 maximum T.I.R and concentric to spindle axis within 0.002 T.I.R.

Carriage alignment—0.002 maximum T.I.R. in 9 inches

Page 6, add the following new paragraph:

3.4.3 *Accuracy.* The brake drum lathe must be capable of truing all brake drums in its drum capacity range in accordance with the following tolerances:

(a) Concentricity—The turned surface of the brake drum shall be within 0.003 inch total indicator reading with axis of hub bore.

(b) Taper—0.001 inch per inch maximum.

(c) Run-out—The turned surface of the brake drum shall not exceed 0.0005 inch T.I.R. reading with axis of workhead spindle.

(d) Final surface—100 microinches, maximum (root mean square averaging method).

Page 6, paragraph 3.5: Add the following new component: "(1) Outboard support (when required)", after component (k).

Page 7, paragraph 3.5.2: Delete "dovetail".

Page 7, paragraph 3.5.2.1: Delete the third and fourth sentences.

Page 7, paragraph 3.5.3.1: Delete the paragraph in its entirety and substitute the following new paragraph:

3.5.3.1 *Spindle drive.* Spindle drive shall be through V-belts, chains, gears, or combinations. Spindle speeds shall be variable through use of stepped pulleys or infinitely variable drive mechanism or quick-change-speed gears. A minimum of three spindle speed ranges shall be provided. Gear drives shall be totally enclosed. Easily adjustable tensioning devices and guards shall be provided where belt or chain drive is employed. All necessary motor mounting brackets, pulleys, belts, chains, sprockets, gears, countershafts, and other parts required to put the lathe in successful operation shall be furnished. Visual means shall be provided to show spindle speeds throughout the entire speed range specified herein.

Page 7, paragraph 3.5.5: Delete the paragraph in its entirety and substitute the following paragraph:

3.5.5 *Tool bar.* The lathe tool bar shall be made from alloy steel and shall be designed to completely preclude a change in rake of the cutting tool bar turning in its clamping device. When specified (see 6.2), the tool bar turning in its clamping device. When specified (see 6.2), the tool bar shall be designed so that it can be rotated 180 degrees in the clamp for turning drums with external braking surfaces.

Page 7, paragraph 3.5.6.1.1: Delete references to "Size 1".

Page 8, paragraph 3.5.6.1.2: Delete the paragraph in its entirety.

Page 8, paragraph 3.5.6.3: Add the following to end of paragraph: "The light shall be equipped with a separate switch that shall function independently of the motor switch."

Page 8, paragraph 3.5.6.4: Delete reference to size I lathe.

Page 8, paragraph 3.5.9: Delete reference to size III lathe.

Page 8, paragraph 3.5.10: Delete the paragraph in its entirety and substitute the following paragraph:

3.5.10 *Accessories.* Unless otherwise specified (see 6.2), all necessary accessories for truing brake drums of military vehicles shall be furnished as well as those accessories required for truing drums of all standard models of United States produced commercial passenger cars. For military vehicles, such accessories shall be provided to fit all bearing cups from the minimum bearing cup MS 19081-58 through the maximum bearing cup MS 19081-173. Floating drum adapters shall be provided for brake drum, Ordnance No. 7328372 and brake drum, Ordnance No. 7025887 as used on military vehicles M38A1 (1/4 ton, 4 by 4 utility truck) and M151 (1/4 ton, 4 by 4 utility truck), respectively. Turning of drums shall be accomplished without chatter.

Page 9: Add the following new paragraph:

3.5.12 *Outboard support.* The use of an outboard support on the size II brake drum lathe shall be optional with the manufacturer. The outboard support shall be a portable type of equipment easily removed for loading and unloading of wheel assemblies from end of spindle.

Page 9, paragraph 3.7: Delete the paragraph in its entirety and substitute the following new paragraph:

3.7 **Fungus control.** When specified (see 6.2), electrical components and circuit elements shall be treated for fungus resistance in accordance with the requirements of MIL-T-152, type II (delete par. 3.1). The windings of the electric motor shall be given two impregnating coats of varnish conforming to type M, grade BB or CB and one sealer coat conforming to type M, grade CA of the requirements of MIL-V-1137 or three impregnating coats of varnish conforming to type M, grade CB, of MIL-V-1137.

Page 9, paragraph 3.10, last sentence: Delete reference to table II and size I.

Page 10, paragraph 4.2, first sentence: Delete "4.6" and substitute "4.5".

Page 10, paragraph 4.2, third sentence: Delete in its entirety.

Page 10, paragraph 4.3.2.1, under major defects: Delete "(AQL shall be 2.5)" and substitute "(AQL shall be 1.5)".

Page 11, under major defects: Delete in its entirety "(AQL shall be 4.0 percent defective)".

Page 11, under major defects: Add new requirement for number 126 as follows: "126. Outboard support not as specified (see 3.5.12)."

Page 11, under major defects: Change old numbers 126 through 131 to new numbers 127 through 132.

Page 11, paragraph 4.3.3, third sentence: Delete "AQL of 1" and substitute "AQL of 4.0".

Page 11, paragraphs 4.4.1, 4.4.1.1, 4.4.1.2 and 4.4.2: Delete in their entirety and substitute:

4.4.1 *Machine accuracy tests.* Each lathe selected for tests shall be tested to determine conformance with machine accuracy requirements specified in 3.4.2. Movable spindles shall be checked at maximum extended position.

4.4.2 *Performance test.* Brake drums with hub, tire and wheel assemblies shall be set up and trued as a typical operation. The concentricity, taper, runout and surface roughness value shall be within the limits specified (see 3.4.3).

4.4.2.1 *Size II lathe.* Unless otherwise specified (see 6.2), the size II lathe selected for tests shall be capable of truing one brake drum of the smallest diameter (8.9 inch minimum diameter), with wheel, tire (7.00-16), hub and drum attached and one brake drum of the larger diameters (17.25 inch maximum diameter) with dual wheels, tires (14.00-24), hub and drum attached."

Page 11: Add the following new paragraphs:

4.4.3 *Spindle speeds.* The lathe shall be tested with a tachometer to determine compliance with spindle speed range and number of speeds required. Failure to comply with the minimum speed range and the minimum speed changes specified shall be cause for rejection (see table I).

4.4.4 *Endurance test.* The size II lathe shall be operated for 200 consecutive start-stop cycles while rotating a 17.25 inch drum with dual wheels, tires (14.00-24), hub and drum attached. The lathe shall be started, brought to a minimum spindle speed of 60 r.p.m., and then allowed to come to a complete stop before starting the next cycle. Upon completion of the 200 cycles, the machine shall be tested in accordance with paragraph 4.4.1. Failure to pass this test shall be cause for rejection. The spindle driving gears shall be visually inspected for overload or fatigue, breakage, cracking, scoring or plastic yielding as defined in ASA Standard B6.12-1954. Any evidence of such defects shall be cause for rejection.

Page 13, paragraph 5.2.1, fifth sentence: Delete reference to size III lathe.

Page 13, paragraph 5.2.2, fourth sentence: Delete reference to size III lathe.

Pages 13 and 14, paragraph 6.2: Delete the paragraph in its entirety and substitute the following paragraph:

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification, also number and date of amendment.

(*b*) Size II—Heavy-duty.

(*c*) Preproduction inspection, when required; also sample quantity, and test report, if different (see 3.1).

(*d*) Safety devices, if different (see 3.3.10).
(*e*) Physical characteristics, if different (see Table I).
(*f*) Tool bar rotation, when required (see 3.5.5).
(*g*) Voltage and current, if different (see 3.5.6.1.1).
(*h*) Cord, when required (see 3.5.6.4).
(*i*) Grinding attachment, when required (see 3.5.8).
(*j*) Crane attachment, when required (see 3.5.9).
(*k*) Accessories, if different (see 3.5.10).
(*l*) Radio interference suppression, if different (see 3.6.1, 3.6.2 and 3.6.3).
(*m*) Fungus control, when required (see 3.7).
(*n*) Painting, if different (see 3.8).
(*o*) Handbook of instructions, if different; and address where additional copies are to be sent (see 3.10).
(*p*) Handbook approval, when required (see 3.10.1).
(*q*) Preproduction inspection, if different (see 4.2).
(*r*) Size of test drum, if different (see 4.4.2.1).
(*s*) Radio interference suppression test agency, if different (see 4.5.1).
(*t*) Level of preservation, packaging, and packing required (see 5.1 and 5.2).
(*u*) Packaging and packing of crane attachment, when required (see 5.1.1.6, 5.2.1 and 5.2.2).

Page 14, paragraph 6.4: Delete the paragraph in its entirety and substitute the following paragraph:

6.4 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Lathes, metal working other than dentists.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Lathes.
Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—WC

Navy—Docks

Air Force—SAA

Preparing activity:

Army—WC